

# **Lyra communication system**

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## Abstract

This report covers the communications system of Munir-3, a CubeSat built by the student organization Lyra and launched from Andøya through NASA's Ghost mission.

The communication system on Munir-3 was developed to test a new antenna concept where the antenna is integrated into the satellite's structural frame.

The system consists of an AVR microcontroller for interfacing with the other subsystems, as well as a TI CC1200 that handles RF generation and the protocols for low-data-rate radio communication at  $433MHz$ . The communication board is supplied with  $5V$  from the satellite's EPS and includes a local regulator that generates  $3.3V$  for the digital components.

The board was produced in two versions: a dedicated transmitter that was integrated into the satellite and launched, and a transceiver version intended as a receiver and for ground testing of the link.

The antenna is a self-designed quarter-wave antenna at  $433MHz$ , implemented with metal wire that folds into a slot at the top of the frame, removing the need for any antenna deployment mechanisms.

The system worked in ground testing, achieving a data rate of up to  $50kb/s$ , and showed no problems during integration and mechanical stress tests. Munir-3 launched as part of NASA's Ghost mission, but we did not receive any data from our system, so we cannot present any flight results.

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# **1 Introduction**

CubeSats place strict constraints on communication system design because of their limited power budget, size, and need for a reliable link to ground within tight mission constraints. The communications subsystem is a key part of the satellite, since it must provide a robust RF link while integrating cleanly with the other onboard electronics and the mechanical structure.

Munir-3 is a 1U CubeSat developed by the student organization Lyra to test a new antenna architecture in which the antenna is integrated into the satellite frame. In the project, the main focus of the communications subsystem was to design a reliable dedicated communications board and antenna integrated into the structural frame.

The PCB was designed around a Texas Instruments CC1200 RF chip responsible for the RF signal generation and signal processing, together with an AVR microcontroller transceiver that handles communications to other subsystems and allows the system to operate as a standalone transceiver. The system receives 5V from the EPS and includes a local regulator to generate 3.3V for the digital components and RF circuitry.

## 2 Theory

The communications subsystem is responsible for establishing and maintaining the radio link between the satellite and the ground station. In a small satellite like a CubeSat, this has to be done within strict limits on power, volume, and mass, while still providing a signal that is strong and clean enough to be received reliably on the ground. A typical communications system consists of a digital part that prepares the data for transmission, an RF front end that converts this data into a modulated radio signal and amplifies it, and an antenna that radiates the signal into space. On the receive side, the same chain operates in reverse: the antenna captures the incoming signal, the RF front end amplifies and down-converts it, and the digital part recovers the data.

The communications subsystem is the direct link between the satellite and ground station and is a critical system for data transmission and mission success. The RF chain must be engineered to produce sufficient output power within power and size limits, maintain an acceptable noise figure, and minimize emissions that could interfere with other systems. At the same time, the antenna has to be physically compatible with the CubeSat structure and dynamics, often without the luxury of large deployables.

In the Munir-3 communications module, the RF chain is built around a Texas Instruments CC1200 transceiver chip, which handles modulation, demodulation, and most of the RF signal generation and processing at 433MHz. An AVR microcontroller connects the communications module to the rest of the satellite, managing configuration of the CC1200, packaging data for transmission, and processing received data. The system is powered from the 5V rail provided by the EPS and uses a local regulator to generate 3.3V for the digital circuitry. Around the CC1200, additional components form the power amplifier (PA) and low-noise amplifier (LNA) stages, impedance-matching networks, and filtering, all of which are needed to ensure efficient transmission.

The antenna on Munir-3 is a custom quarter-wave monopole at 433MHz that is integrated directly into the satellite's structural frame. Instead of using a deployable boom or external whip, a metal element is folded into a slot at the top of the frame. This integration affects the antenna's impedance, radiation pattern, and bandwidth, and therefore has to be considered in both the mechanical and RF designs. The following subsections describe the theoretical background for the antenna and the main elements of the communications PCB in more detail.

### 2.1 Antenna

An antenna converts electrical signals in a transmission line into electromagnetic waves that propagate through space, and vice versa on receive. For a monopole antenna operating over a ground plane, the antenna length is often chosen to be a quarter of the wavelength at the

operating frequency. At 433MHz, the free-space wavelength is:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{433 \times 10^6} \approx 693mm$$

So a quater wave antenna would is  $\frac{\lambda}{4} = \frac{693}{4} \approx 173mm$ . This antenna fits within the physical constraints of a 1U CubeSat while still providing reasonable gain and bandwidth.

The performance of a real antenna differs from the ideal case because of nearby structures, materials, and the finite size of the ground plane. Integrating the antenna into the satellite frame means that the frame itself becomes part of the antenna system. Metal parts, fasteners, and other subsystems close to the antenna can shift its resonant frequency, change the input impedance seen by the RF front end, and distort the radiation pattern. To ensure efficient power transfer, the antenna impedance must be matched to the output of the transceiver and PA through matching networks, and the structural environment needs to be considered when estimating gain and polarization.

On receive, the same factors determine how much signal power the antenna captures from an incoming wave. Any mismatch or loss in the antenna and its feed line reduces the effective sensitivity of the communications system, which is particularly critical for low-power CubeSat links.

For larger satellites, the communications subsystem is often divided into two distinct parts. One handles housekeeping and status data using a low data rate, lower transmit power, and typically an omnidirectional antenna to maintain a robust link regardless of spacecraft attitude. The other is dedicated to payload data and is designed for much higher throughput, using directional antennas and higher-power transmitters to achieve a high-speed link to the ground. In 1u cubesats you usually only have the budget for one system where you will usually use a more reliable system with an omnidirectional antenna.

## 2.2 PCB for communications

The communications PCB implements the complete RF and digital signal chain between the satellite bus and the antenna. It hosts the CC1200 transceiver, the AVR microcontroller, the local 3.3V regulator, and the discrete components that form the PA, LNA, matching networks, and filters. The PCB layout is a crucial part of the RF design: trace geometry, ground planes, and component placement all influence parasitic capacitances and inductances, which in turn affect impedance matching, spurious emissions, and overall stability. Almost all RF components are matched to 50Ω. This is a widespread impedance for RF matching it makes matching to other components simple.

At the system level, the PCB has to provide a clean interface to the rest of the satellite, which includes power from the EPS, digital communication with the onboard computer, and defined RF connections to the antenna integrated in the frame. It is important to separate noisy digital circuitry from sensitive RF paths by using proper decoupling, controlled-impedance traces, and

dedicated ground return paths. The following subsections outline the theory behind the main functional blocks on the PCB: RF generation in the CC1200, the power amplifier stage, and the low-noise amplifier on receive.

### **2.2.1 RF generation**

RF generation in the Munir-3 communications system is centered on the TI CC1200 transceiver. Internally, the CC1200 uses a frequency synthesizer based on a phase-locked loop (PLL) to generate a stable carrier at the desired operating frequency, in this case 433MHz. The PLL locks a voltage-controlled oscillator (VCO) to a reference crystal, allowing precise control over the output frequency and supporting different modulation schemes such as frequency shift keying (FSK) or Gaussian FSK.

Digital data from the AVR microcontroller is converted into a modulated RF signal inside the CC1200 by varying the instantaneous frequency or phase of the carrier according to the chosen modulation format. The output of the CC1200 passes through matching networks that transform its internal output impedance to  $50\Omega$ , matching the rest of the RF transmission lines. These networks, together with any external filtering components, help reduce reflections, minimize out-of-band emissions, and ensure that as much of the generated RF power as possible reaches the antenna and gets propagated into space.

### **2.2.2 PA**

The power amplifier (PA) stage boosts the RF signal from the transceiver to the desired output power level for transmission. In a CubeSat, the PA has to balance several competing requirements: it must provide enough gain and output power to close the link budget to ground, while remaining efficient enough to operate within the satellite's limited power budget and thermally stable under vacuum and varying temperatures.

### **2.2.3 LNA**

On the receive side, the low-noise amplifier (LNA) is the first active component after the antenna and matching network. Its main purpose is to amplify weak incoming signals while adding as little noise as possible. The noise figure of the LNA has a dominant impact on the overall sensitivity of the receiver, because noise introduced at this stage is amplified by all subsequent stages.



### 3 Method and design

The Munir-3 communications subsystem was designed to provide a reliable low-data-rate link using an omnidirectional antenna, while staying within the strict power and volume constraints of a 1U CubeSat. The design centered on two main elements: the frame-integrated antenna and the dedicated communications PCB, together with their interfaces to the EPS and the onboard computer.

#### 3.1 Antenna

##### 3.1.1 Design

The antenna is implemented as a quarter-wave monopole at  $433MHz$  integrated directly into the satellite frame. Based on the wavelength calculation from the theory section, the target physical length of the monopole is approximately  $173mm$ , which has to be implemented within the dimensions of a 1U CubeSat. The radiating is implemented as a wire folded into a dedicated slot at the top of the frame. This element is mechanically fixed to the structure and electrically connected to the communications PCB. Figure 1 shows the antenna integrated into the rail in the top frame.

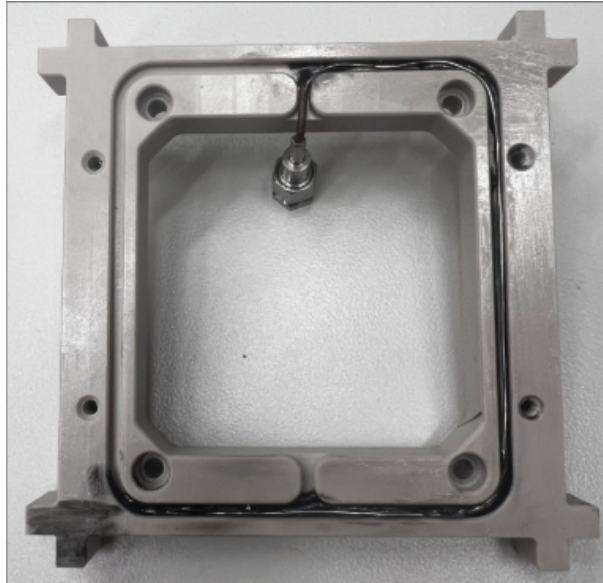


Figure 1: Integrated antenna

Integrating the antenna into the frame removes the need for separate antenna deployment mechanisms and ensures that the antenna remains rigidly attached during launch and operation. However, because the antenna shares structure with the frame and other metallic components, its exact impedance and radiation pattern differ from the ideal quarter-wave monopole. To mitigate these effects, the frame was made of a non-conductive material. The antenna feed is connected through a  $50\Omega$  matching network at  $433MHz$  to ensure the antenna radiates as much of the signal as possible.

### 3.1.2 Test setup

For initial antenna testing we used a liteVNA. This instrument tells us the frequency response and S-parameters of our antenna. The goal was to confirm the transmitted frequency, estimate the effective radiated power, and check the impedance matching at  $433\text{MHz}$ .

These tests were used to show that the frame-integrated antenna, together with its matching network, could support the link we wanted and was mechanically robust through handling and integration. The antenna configuration was then used unchanged for the subsequent system-level tests and vibration campaigns.

## 3.2 PCB for communications

### 3.2.1 Design

The communications PCB implements the complete RF and digital signal chain between the satellite bus and the antenna. It utilizes a TI CC1200 transceiver, an AVR microcontroller, a local  $3.3\text{V}$  regulator, and discrete components that form the PA, LNA, matching networks, and filters.

The CC1200 was selected because it provides a flexible, low-power RF front end with integrated modulation, demodulation, and packet handling at  $433\text{MHz}$ , as well as its ease of use. The AVR microcontroller was chosen as a simple and well-understood platform for configuring the CC1200, interfacing with other subsystems, and managing packet formatting. This combination allowed the communications board to operate both as an integrated flight transmitter and as a standalone transceiver as a ground station. Figure 2 shows the transceiver PCB.

The PCB receives  $5\text{V}$  from the EPS and uses an onboard linear regulator to generate  $3.3\text{V}$  for the CC1200 and the AVR. Locally regulating down to  $3.3\text{V}$  provides a stable supply voltage, reduces noise coupling from the  $5\text{V}$  bus, and matches the operating requirements of the RF and digital components. Input and output decoupling capacitors are placed close to the regulator pins to stabilize the supply and reduce high-frequency noise on the power rails.

The RF path on the PCB is implemented using controlled-impedance traces and matching networks designed for  $50\Omega$ . The CC1200 RF input and output pins are connected to an impedance-matching network composed of inductors and capacitors, chosen to transform the internal impedance of the chip to  $50\Omega$ .

On the transmit side, the CC1200 generates the modulated RF signal at  $433\text{MHz}$  using its internal frequency synthesizer and modulator. The output passes through a matching network and a power amplifier stage designed to provide sufficient output power to close the link budget to the ground station. The PA amplifies the signal and additional filtering is used to suppress harmonics and out-of-band emissions.

On the receive side, the antenna signal is fed through a matching network into a low-noise amplifier (LNA) designed for  $50\Omega$  input impedance. The LNA boosts weak incoming signals

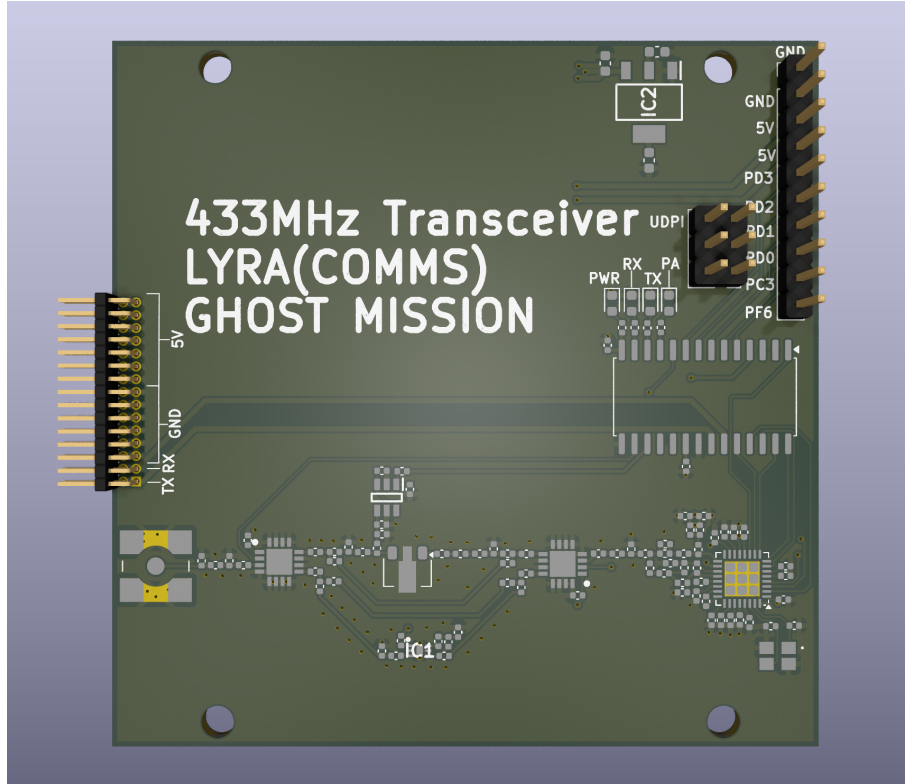


Figure 2: Transceiver PCB

before they enter the CC1200 receiver front end, improving overall sensitivity. The LNA biasing and matching network were chosen to achieve a low noise figure and stable operation while remaining compatible with the board power budget.

At the system level, the communications PCB interfaces to the rest of the satellite through the backplane. Power is supplied from the EPS 5V rail via dedicated connectors, and digital communication with the onboard computer is achieved using simple serial interfaces(UART in our case). Care was taken in the layout to separate noisy digital traces from sensitive RF paths, using ground planes, decoupling capacitors, and short ground return paths to reduce noise coupling.

### 3.2.2 Experimental setup

For bench testing, the communications PCB was first evaluated in isolation. The board was powered from a laboratory power supply set to 5V, replicating the EPS output, and the onboard 3.3V regulator output was verified using a multimeter. The CC1200 and AVR microcontroller were programmed via a UPDI debug interface, and basic transmit and receive functionality was tested using a dedicated ground station and an off-the-shelf antenna.

## 3.3 System testing

For the system testing, the communications PCB was installed in the satellite structure, and the RF feed was connected to the antenna. Tests used simple end-to-end link measurements

between the flight transmitter board mounted in the satellite and a ground station. These tests verified that the integrated antenna could successfully radiate and receive signals when operating as part of the complete structure.

## 4 Results

This section summarizes the measured performance of the Munir-3 communications subsystem, covering the communications PCB, the frame-integrated antenna, and system-level tests with the rest of the satellite. As with the EPS and firmware subsystems, all results are based on pre-flight, ground-based testing. No communications data were received after launch, so in-orbit performance could not be verified.

### 4.1 Communications PCB performance

The communications PCB was first evaluated in isolation. During these tests, the board was powered from a laboratory supply set to  $5V$  to replicate the EPS output, and the onboard  $3.3V$  regulator output was verified with a multimeter before RF measurements. The AVR microcontroller and CC1200 transceiver were programmed via a UPDI interface, and basic transmit/receive operation was confirmed.

The CC1200 was configured to operate at a center frequency of approximately  $433MHz$ , using the default modulation and packet settings for the CC1200. The nominal RF output power at the CC1200 pins was set to about  $+10dBm$ . A dedicated power amplifier stage on the PCB was designed to provide an additional gain of roughly  $20dB$ , giving an intended total output power on the order of  $+30dBm$  at the antenna feed, subject to losses in the matching network and feed line.

In transmit mode at this configuration, the measured current draw from the  $5V$  supply was approximately  $250mA$ . This current includes the consumption of the CC1200, the AVR, and the PA.

Short-range link tests using the PCB and a dedicated ground receiver were performed at different data rates. At a distance of about  $10m$  indoors and a data rate of  $50kb/s$ , the received signal level was around  $-35dBm$ . At the same distance but with a lower data rate of approximately  $1.2kb/s$ , the received power improved to about  $-27dBm$ . Both configurations achieved reliable packet reception over these short ranges, but the received power levels indicate that the overall RF path, including matching networks and feed, was not completely lossless. Some inefficiencies in the PA, impedance matching, and antenna coupling are likely contributors to the observed difference between the intended and received signal strength.

No hardware faults or critical issues were identified on the communications PCB during these bench tests. The board successfully transmitted and received packets using the CC1200 and AVR combination, and the RF chain behaved stably across the tested configurations.

## 4.2 Antenna performance

The frame-integrated antenna was characterized separately before full system integration. The antenna is a quarter-wave monopole at  $433\text{MHz}$  realized as a metal wire folded into a slot at the top of the satellite frame, connected to the PCB through a  $50\Omega$  matching network. Because the antenna is part of the CubeSat structure and is surrounded by other mechanical and electrical components, its actual impedance and radiation pattern differ from the ideal, freestanding monopole assumptions used in the theory section.

### 4.2.1 Impedance and matching

The impedance of the integrated antenna was evaluated using a vector network analyzer connected at the feed point. Measurements of the reflection coefficient  $S_{11}$  around  $433\text{MHz}$  showed values in the range of approximately  $-10$  to  $-12\text{dB}$  at the target frequency. This level of return loss corresponds to a moderate match: about 90% of the transmit power is delivered to the antenna, but the rest is still reflected back toward the PA.

The measured bandwidth over which  $S_{11}$  remained close to these values was sufficient to cover the intended operating channel.

### 4.2.2 Radiation behaviour and link tests

Practical link tests with the frame-integrated antenna installed on the satellite structure were performed in parallel with the PCB tests. At distances on the order of  $10\text{m}$  indoors, the system achieved reliable communication at both  $50\text{kb/s}$  and  $1.2\text{kb/s}$ , with the received power levels mentioned above. These measurements are consistent with the moderate impedance match and with the presence of losses in the feed and matching network.

The antenna appeared to behave approximately omnidirectionally. Rotating the satellite during tests did not reveal strong nulls or distinctly directional lobes in the received signal, and there was no clear evidence of significant gain in any particular direction. Within the constraints of the test setup, the antenna behaved much like a simple monopole with relatively uniform coverage.

The mechanical robustness of the antenna was also verified. The antenna configuration was kept unchanged for subsequent integration and vibration testing, and no damage or electrical discontinuity was observed. After vibration tests, the communications system continued to operate as before, suggesting that the antenna mounting and feed-through are mechanically sound.

### 4.3 System testing

System-level tests were carried out with the communications PCB installed in the satellite frame, connected to the EPS, the onboard STM32-based computer, and the frame-integrated antenna. In this configuration, the data flow followed the intended flight architecture: the STM32 collected sensor and payload data, packaged the measurements into telemetry packets, and forwarded them over a serial link to the AVR on the communications board. The AVR then programmed the CC1200, wrote the packet payload into the transmit FIFO, and initiated RF transmission.

On the ground, a second CC1200- and AVR-based receiver was used as a dummy ground station. This receiver captured the transmitted packets and forwarded the data to a computer for storage and analysis. In this setup, end-to-end tests showed that telemetry packets from the STM32 could be successfully transmitted through the onboard communications chain, received by the ground station hardware, and decoded correctly.

These tests demonstrated that the complete communications path—from data collection on board to reception on the ground—worked as intended under laboratory conditions. The system operated reliably at data rates up to about  $50\text{kb/s}$  over short distances, with no obvious packet loss within the scope of the tests. This is consistent with the standalone PCB and antenna performance measurements.

The communications subsystem also took part in mechanical stress and vibration testing alongside the other satellite subsystems. Before and after vibration tests, functional checks confirmed that the communications PCB and antenna still transmitted and received packets as expected. No connector issues, solder failures, or obvious RF path damage were observed, indicating that the mechanical design of the communications subsystem is robust enough to survive launch-like vibration.

### 4.4 Limitations of the results

A key limitation of the communications results is that they are based entirely on pre-flight, ground-based testing. Munir-3 was launched as part of NASA's GHOST mission, but no communications data were received after launch. As a result, there is no way to confirm how the communications subsystem behaved in orbit, nor to determine whether any fault occurred in the RF chain, the antenna, the EPS, or elsewhere in the satellite.

The ground tests themselves were also limited. They did not fully replicate the conditions of space, such as vacuum and radiation. Vibration tests provided confidence that the hardware could survive launch, and short-range link tests showed that the communications design worked in the lab, but these conditions are not a complete substitute for a space environment.

Results show that the communications subsystem—consisting of the AVR and CC1200-based PCB and the integrated antenna correctly on the ground and through mechanical testing. At the same time, the lack of any received flight data leaves the final behaviour of the system under real

mission conditions as an open question, and points to the need for further work on end-to-end testing and flight verification in future iterations.



## 5 Discussion

The Munir-3 communications subsystem was designed as a simple, low-data-rate link built around a CC1200 transceiver, an AVR microcontroller, and an antenna integrated into the satellite frame. The results show that this combination works under laboratory conditions and through vibration testing, but also that the RF chain and antenna are not perfectly efficient.

### 5.1 PCB and RF performance

The communications PCB operated as intended at  $433\text{MHz}$ , with the CC1200 and PA supporting data rates up to  $50\text{kb/s}$ . The measured current draw of about  $250\text{mA}$  at  $5\text{V}$  is reasonable for a small amplifier chain, and short-range tests over  $10\text{m}$  showed stable packet reception.

At the same time, received levels of  $-35\text{dBm}$  at  $50\text{kb/s}$  and  $-27\text{dBm}$  at  $1.2\text{kb/s}$  are lower than would be expected from an ideal  $+30\text{dBm}$  link at such short ranges. This points to losses in the PA, matching network, feed line, and antenna, meaning more iterations and more thorough testing are needed to enhance the system further.

### 5.2 Antenna integration and matching

Integrating the antenna into the frame removes the need for a separate deployment mechanism and ties the antenna directly to the structure. The measured  $S_{11}$  between  $-10$  and  $-12\text{dB}$  at  $433\text{MHz}$  shows a usable match to  $50\Omega$ , but this could definitely be improved. The link measurements are consistent with this moderate match and suggest that some transmit power is reflected rather than radiated.

The antenna fits within the frame, appears approximately omnidirectional, and survived vibration tests without damage. Future iterations could benefit from more detailed antenna characterization and matching optimisation to increase the link margin.

### 5.3 System-level behaviour and missing flight data

System-level tests confirmed that the communications subsystem integrated properly with the EPS and firmware. Telemetry packets collected by the STM32, passed through the AVR, and transmitted by the CC1200 were received and decoded correctly by the dummy ground station. The subsystem also remained functional before and after vibration testing, indicating that the mechanical design and interfaces are robust.

The main limitation is the complete absence of flight data. Everything here rests on pre-flight, ground-based testing, and no data came back after Munir-3 was launched. This makes it impossible to say whether the communications subsystem failed in orbit, whether another subsystem prevented it from transmitting, or whether external factors were responsible. Understanding this

gap requires looking at the full chain, from the STM32 and EPS through the radio and antenna to the ground station.

## 5.4 Lessons for future work

The communications results suggest several points for future missions:

- Frame-integrated antennas are feasible and can support a functional link, but their matching and radiation must be treated as part of the whole structure, not as ideal standalone components.
- Modest inefficiencies in the PA and matching network can significantly reduce effective radiated power, so careful RF layout and combined testing are important.
- Ground tests give reasonable confidence in the design, but they cannot replace in-orbit verification.

Overall, the subsystem met its goals on the ground and through mechanical testing, but the lack of flight data leaves the true in-orbit behaviour of the communications system as an open question and highlights the need for more margin and more exhaustive end-to-end testing in future CubeSat projects.

## 6 Conclusion

The Munir-3 communications subsystem was designed to provide a simple, robust, low-data-rate link using a CC1200 transceiver, an AVR microcontroller, and a frame-integrated quarter-wave antenna at  $433\text{MHz}$ . Ground-based results show that this architecture works as intended. The communications PCB operated correctly at  $433\text{MHz}$ , supported data rates up to  $50\text{kb/s}$ , and integrated cleanly with the EPS and onboard STM32-based computer. End-to-end tests demonstrated successful transmission and reception of telemetry packets through the complete chain, and vibration testing confirmed that the mechanical design of the PCB and antenna is robust.

At the same time, the measured received power levels and antenna  $S_{11}$  indicate that the RF chain is not fully efficient. Losses in the PA, matching network, feed line, and integrated antenna reduce the effective radiated power compared to the ideal design. The frame-integrated antenna provides a usable, approximately omnidirectional pattern within the constraints of a 1U CubeSat, but its matching and radiation characteristics leave room for improvement in future iterations.

The largest limitation is the absence of any flight data from Munir-3. All conclusions in this report are based on pre-flight, ground-based testing, and no telemetry was received after launch. As a result, the true in-flight behaviour of the communications subsystem, and of the satellite as a whole, remains unknown. It is not possible to determine whether the radio, the antenna, the EPS, or another subsystem prevented successful communication.

Overall, the work shows that a frame-integrated antenna and a compact CC1200/AVR communications PCB can provide a functional link for a small CubeSat, and that the design survives mechanical testing. Future missions can build on this by improving RF efficiency and antenna matching, adding diagnostic modes and redundancy, and performing more exhaustive end-to-end tests under conditions that more closely resemble the space environment.

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